Introduction to Clustering

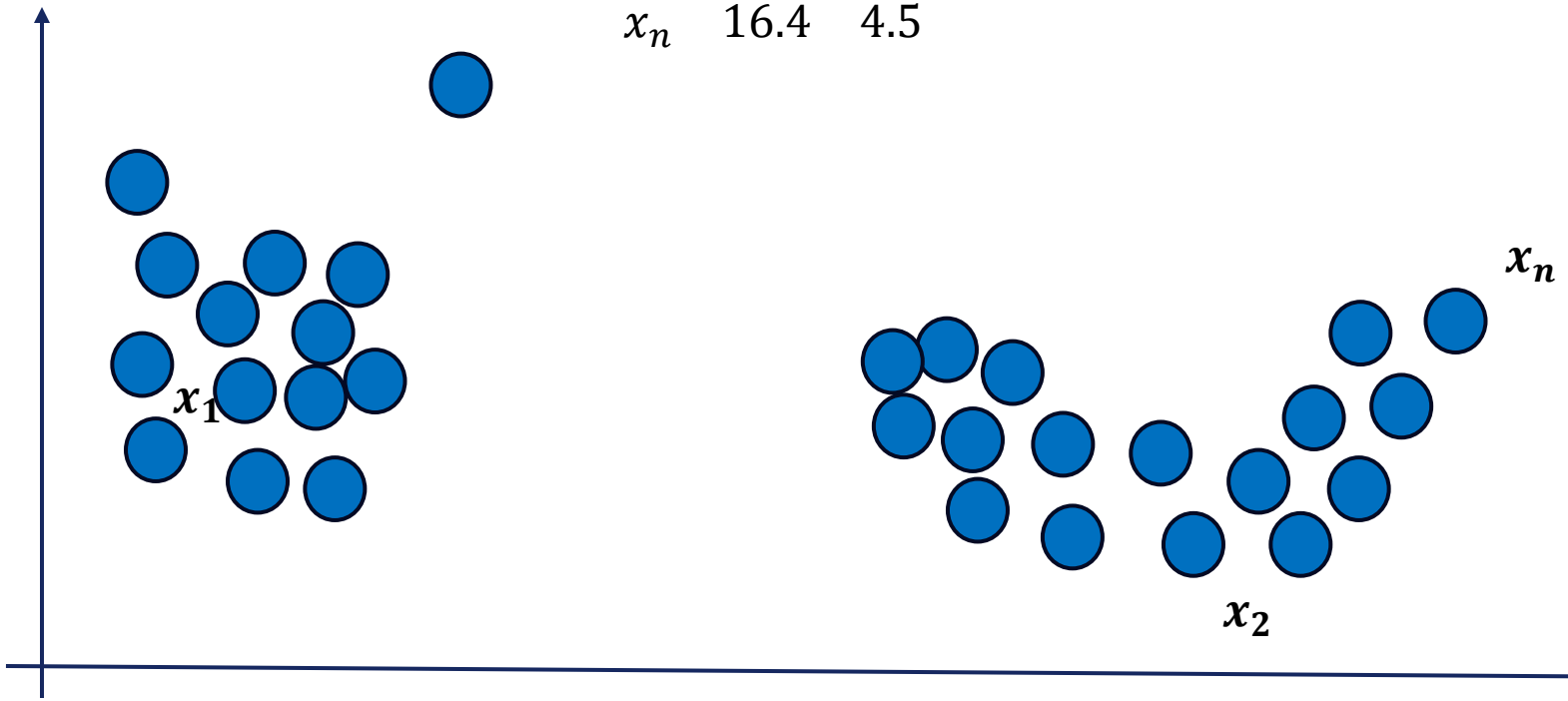
CSCI 347

Adiesha Liyana Ralalage

What are clusters in a dataset

Data Matrix

	X_1	X_2
x_1	2	3.2
x_2	13.1	0.9
$\vdots$	$\vdots$	$\vdots$
x_n	16.4	4.5



Clustering

- Clustering is the task of partitioning the points into natural groups.
- These natural groups are called clusters.
- Points within these clusters are very similar, whereas points across clusters are dissimilar as possible.

Applications of clustering

Data Matrix

Movies

People

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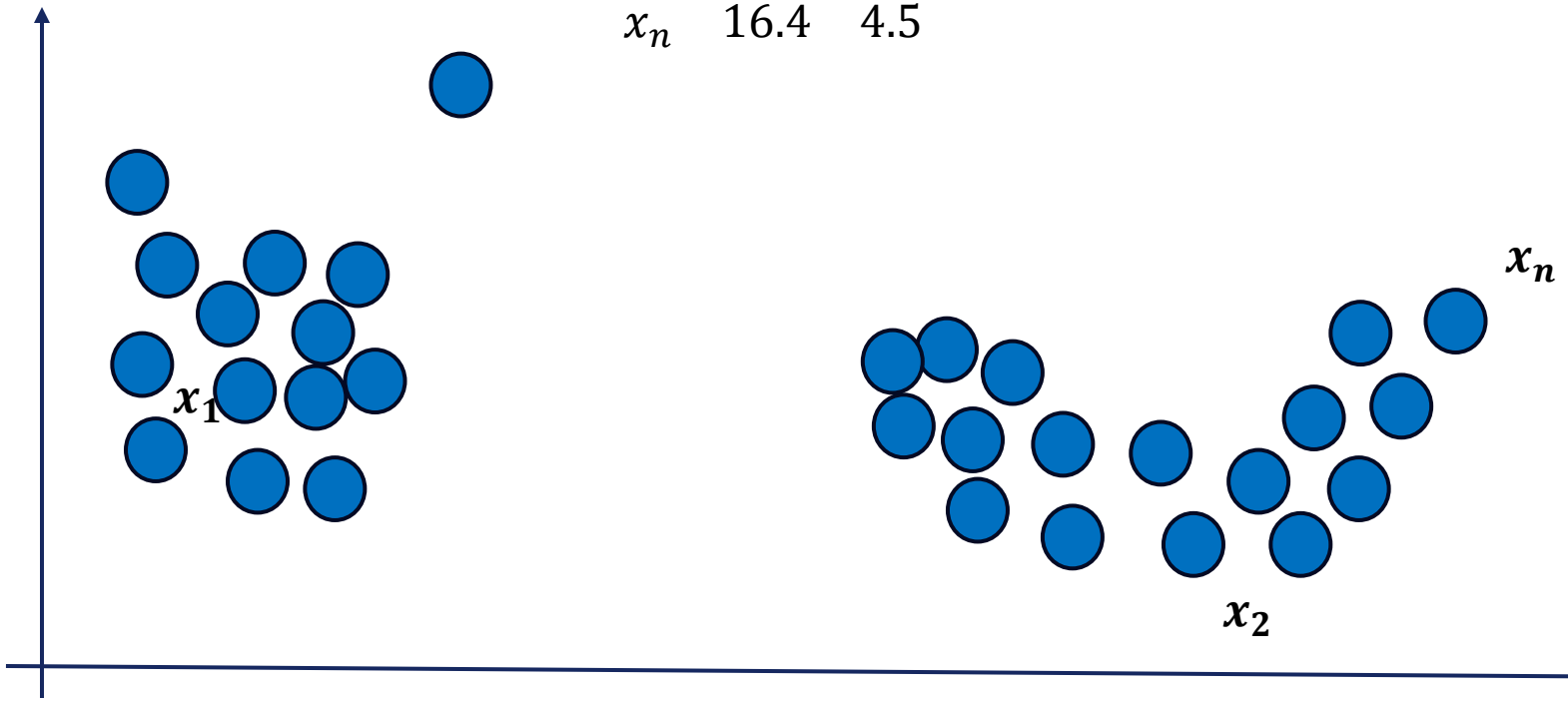

Applications of clustering

Data Matrix

Samples

Genes

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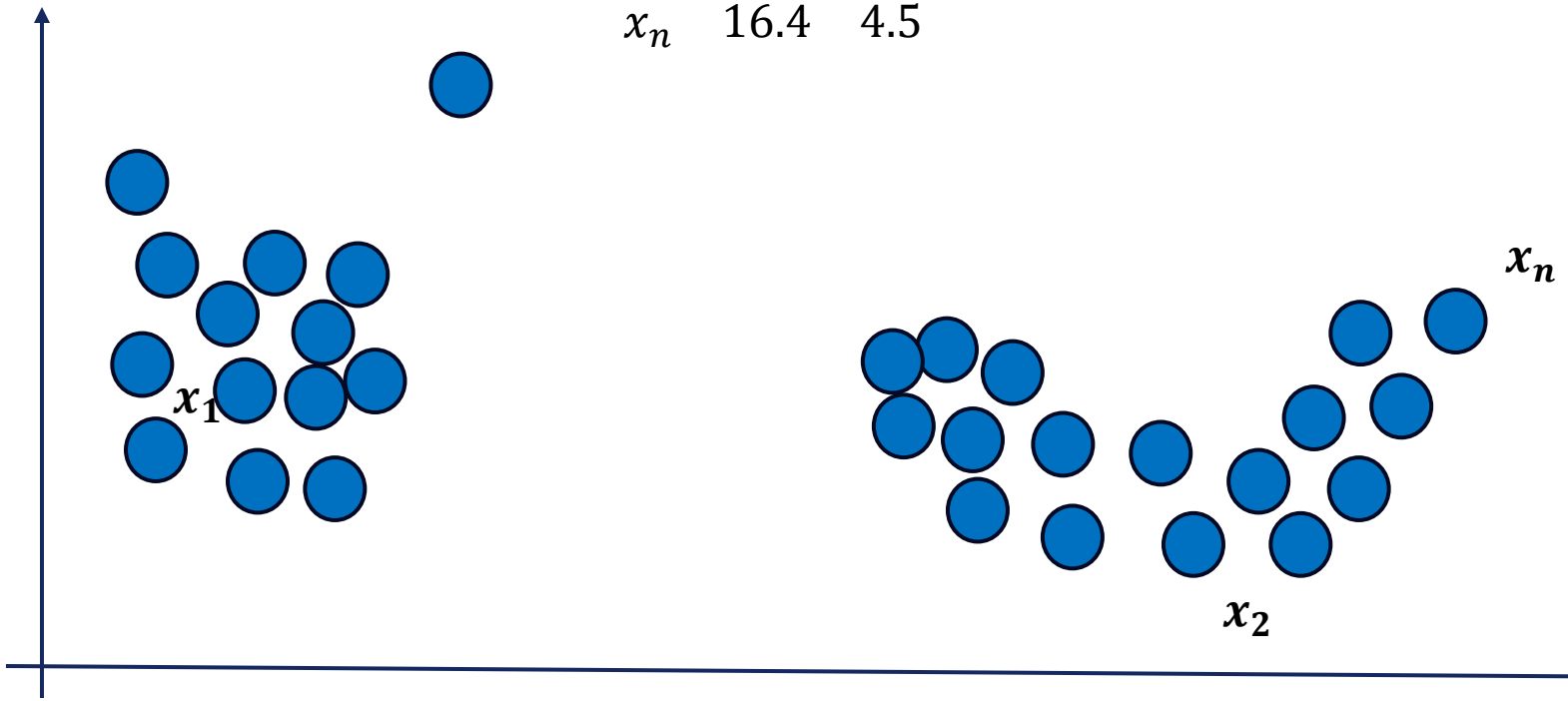

Applications of clustering

Data Matrix

Words

Documents

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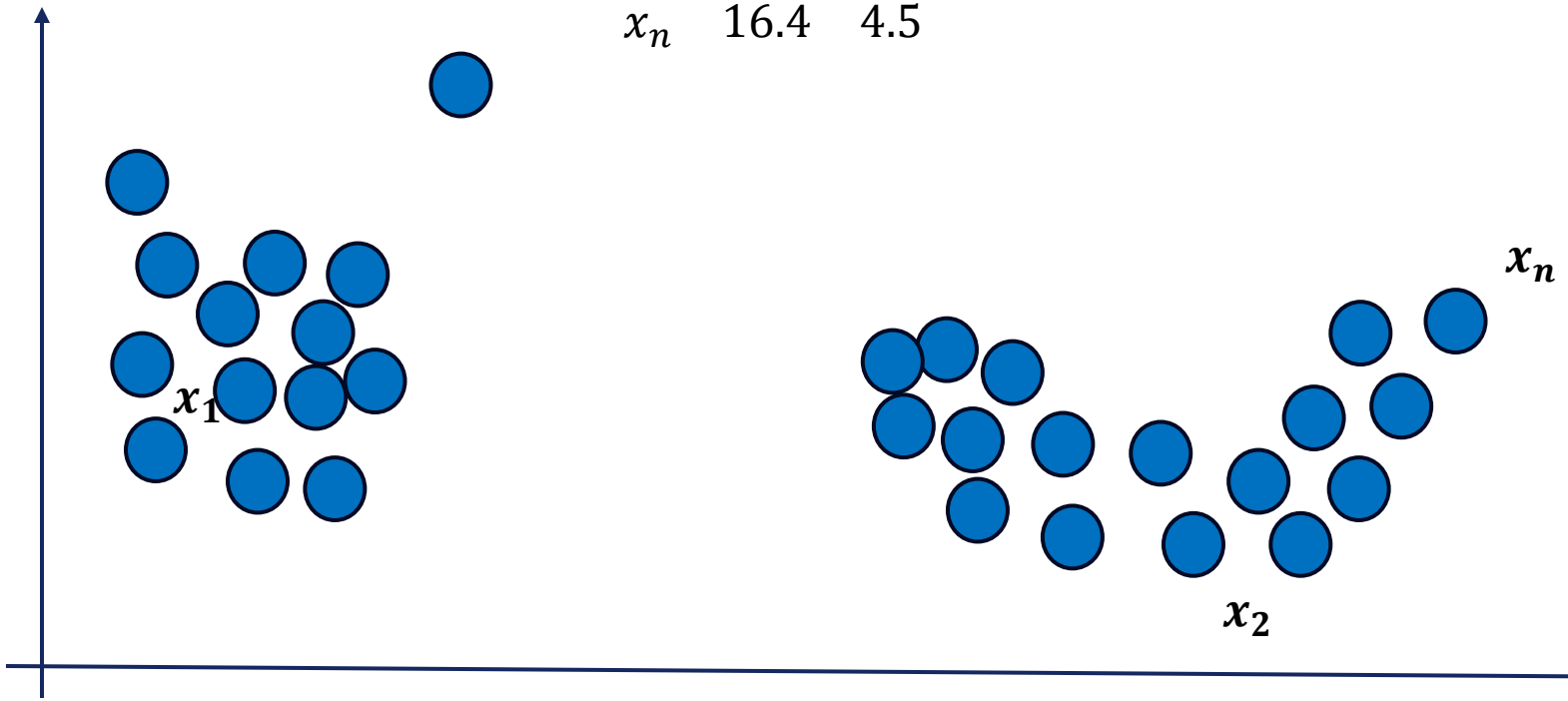

How do we find clusters in a dataset?

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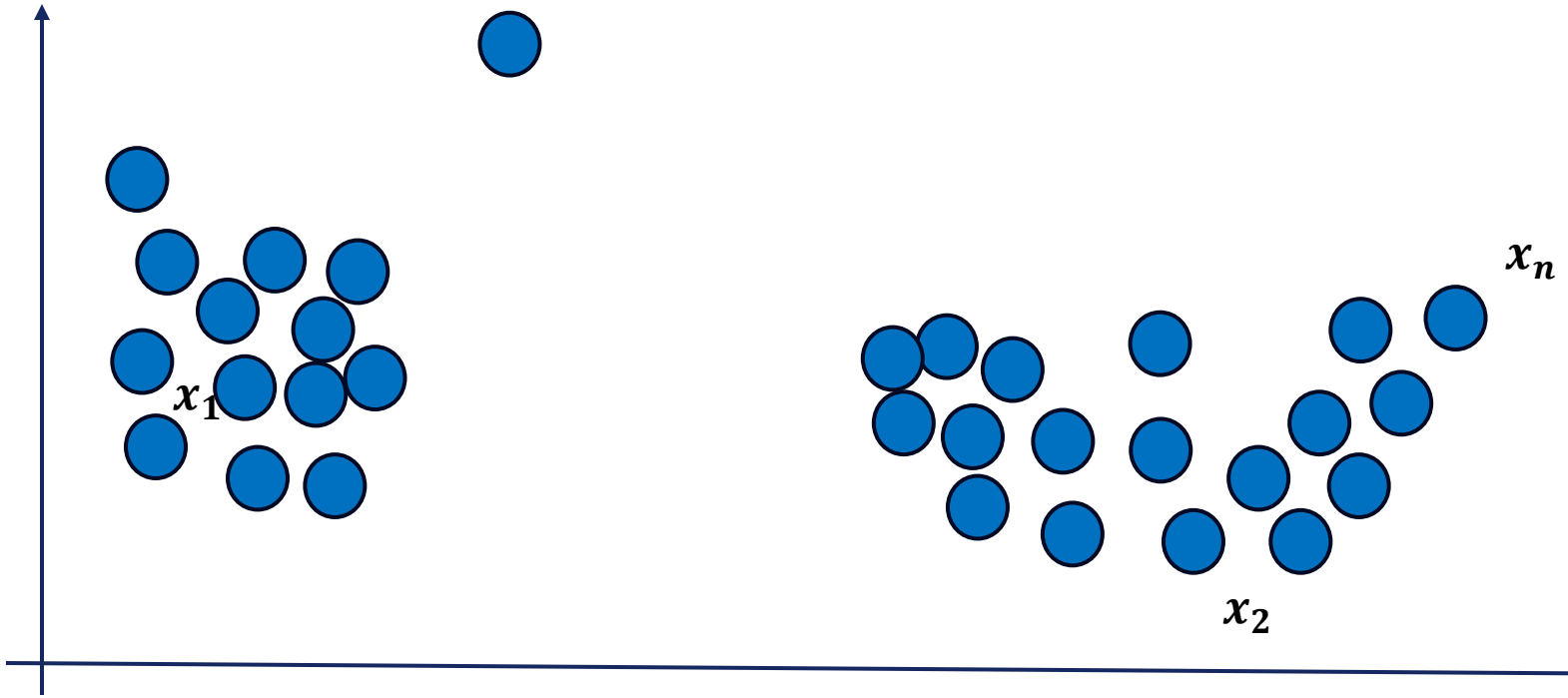
Our goal is to gather data instances into groups with high within-group similarity



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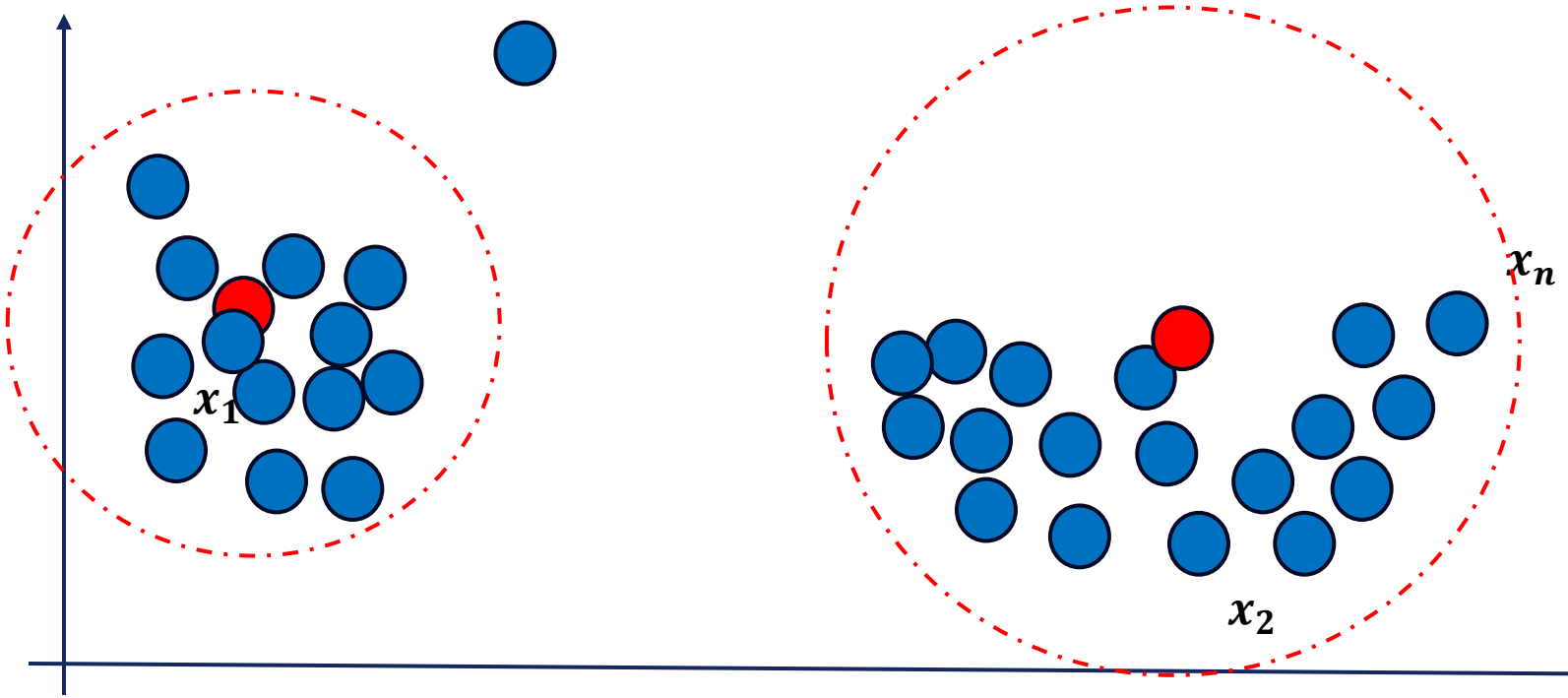
Representative-based methods

Find a representative that best represents each cluster, and group points based on their closest representative.



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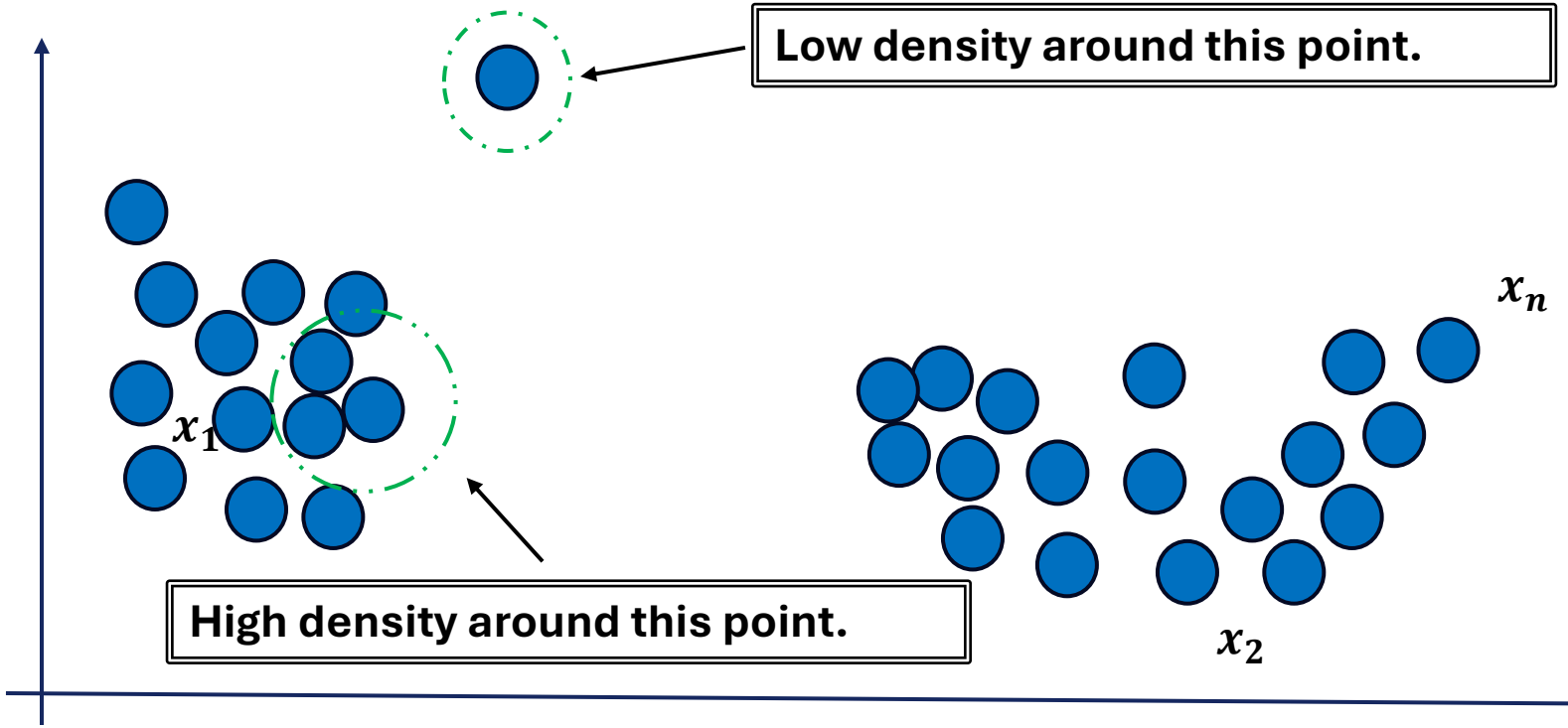
Density-based methods:

Find regions of high density (# points / some small volume)



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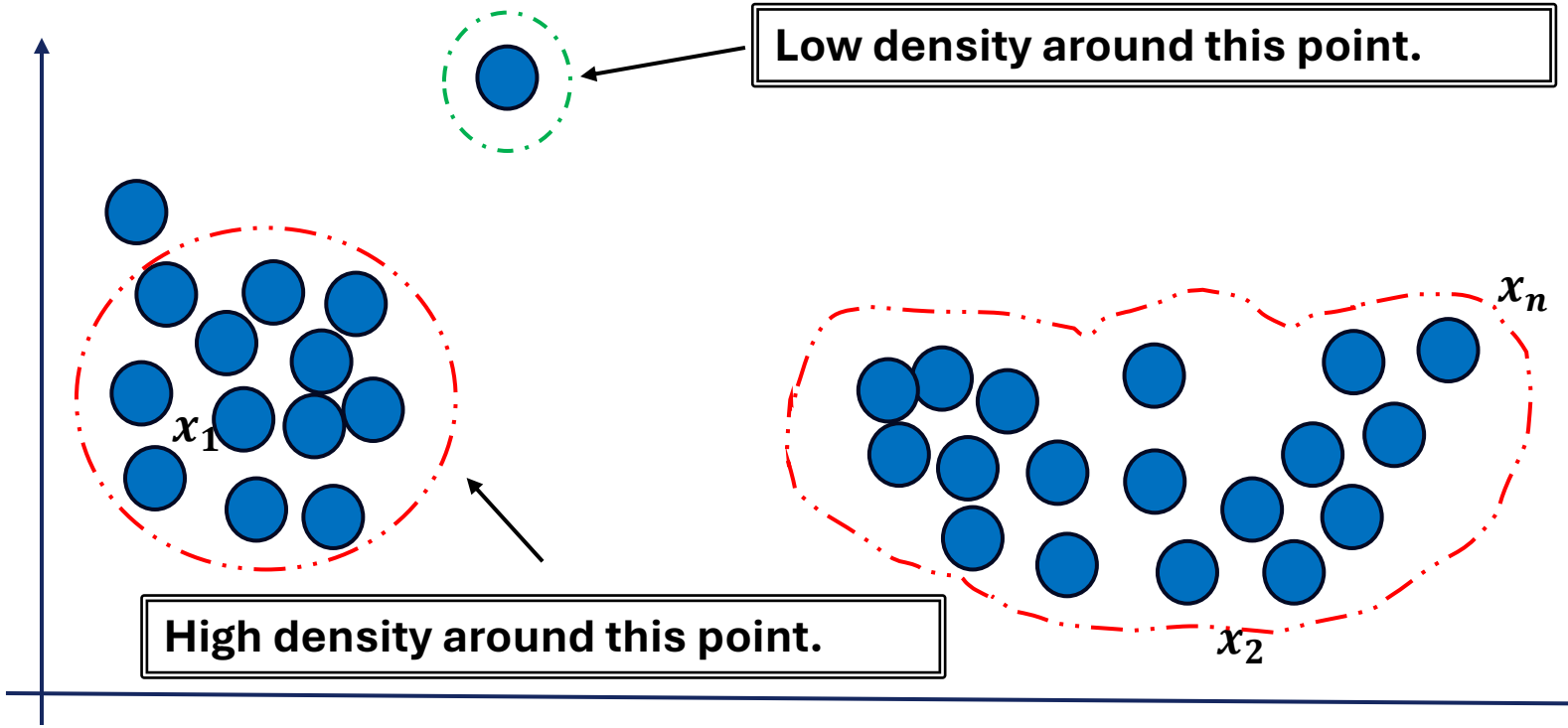

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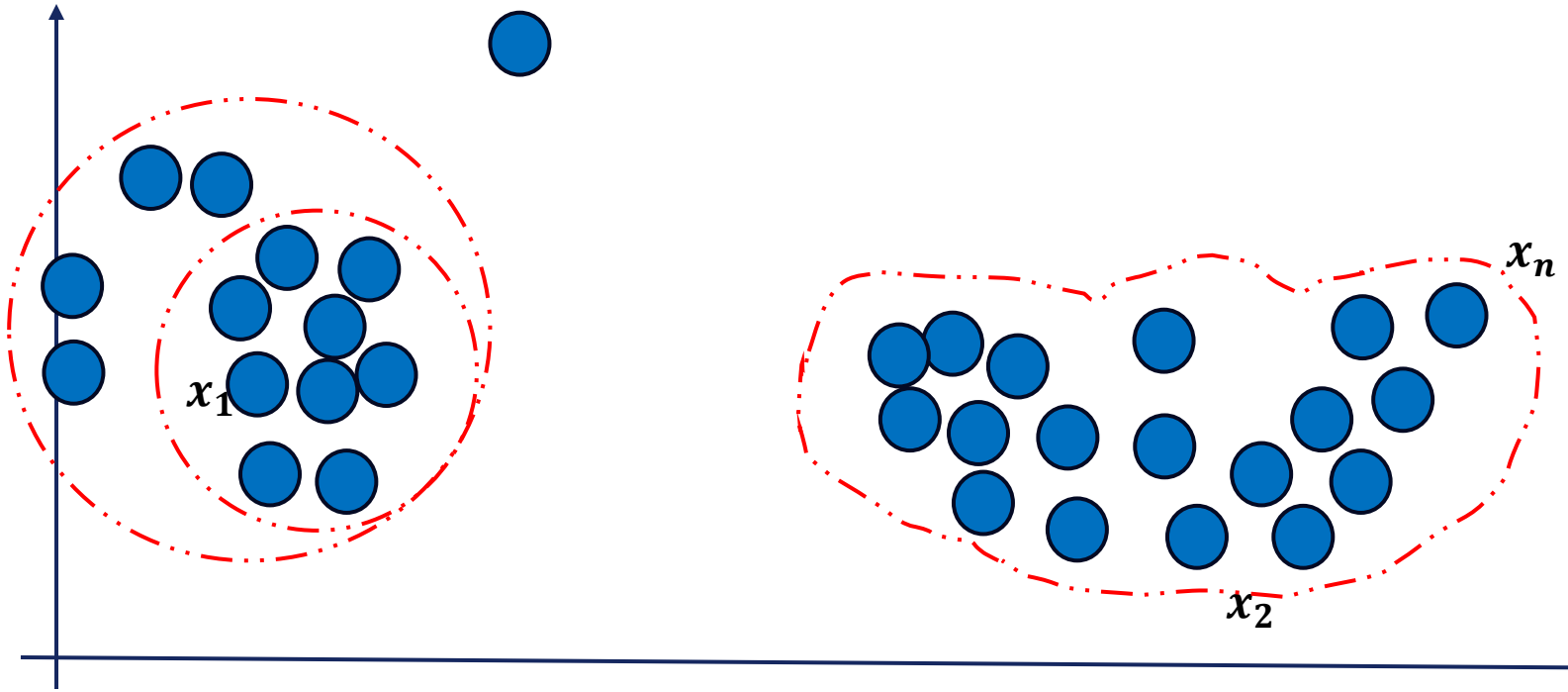
Hierarchical methods:

Clusters within clusters

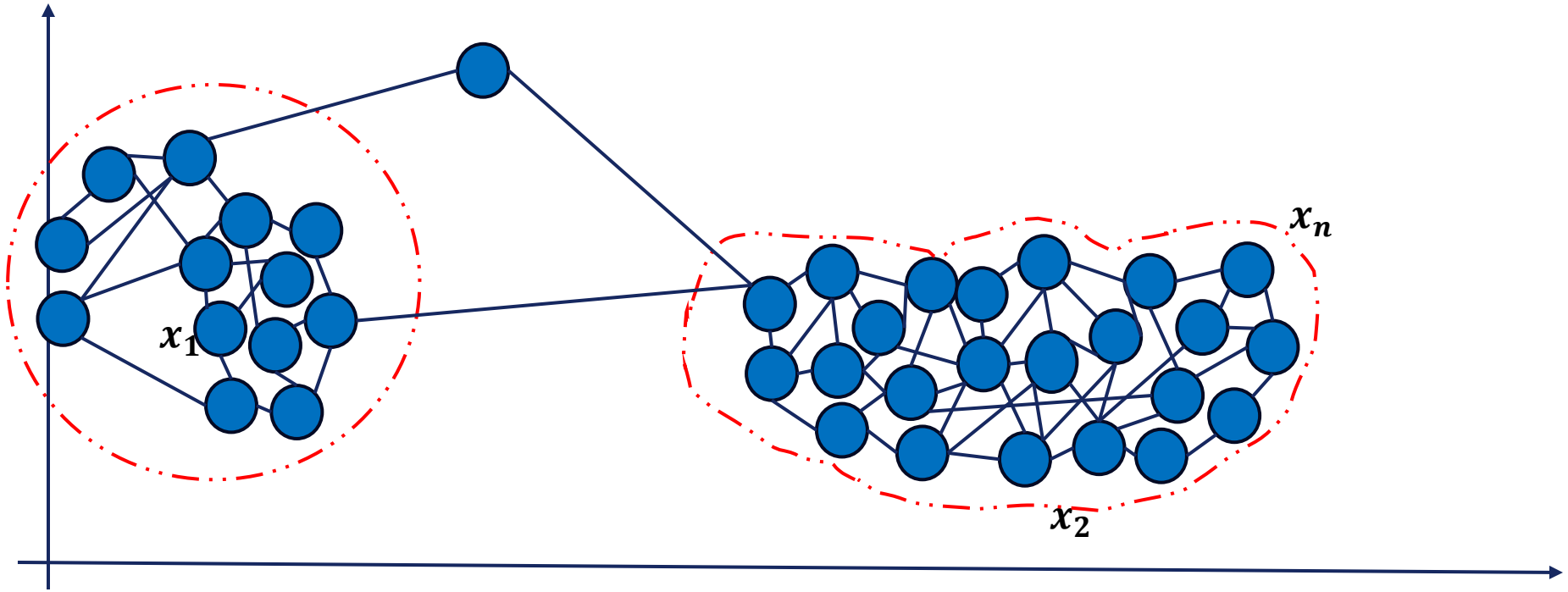


How do we find clusters in a dataset?

$$\begin{array}{ccc} & X_1 & X_2 \\ x_1 & 2 & 3.2 \\ x_2 & 13.1 & 0.9 \\ \vdots & \vdots & \vdots \\ x_n & 16.4 & 4.5 \end{array} \longrightarrow \begin{array}{ccccc} & x_1 & x_2 & \cdots & x_n \\ x_1 & 0 & 0 & \cdots & 0 \\ x_2 & 0 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n & 0 & 0 & \cdots & 0 \end{array}$$

Graph based methods:

Find subgraphs with high edge connectivity



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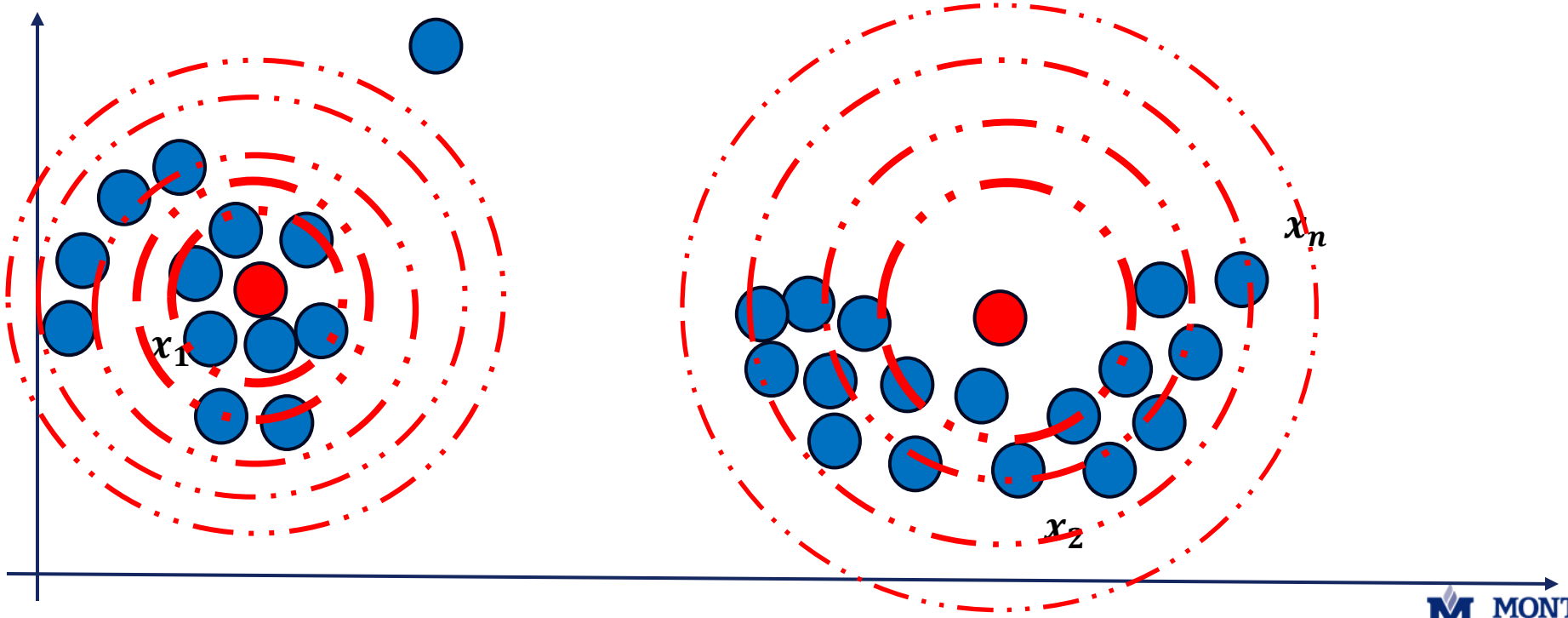
Soft clustering or probabilistic clustering

Estimate the probability distribution that the points come from



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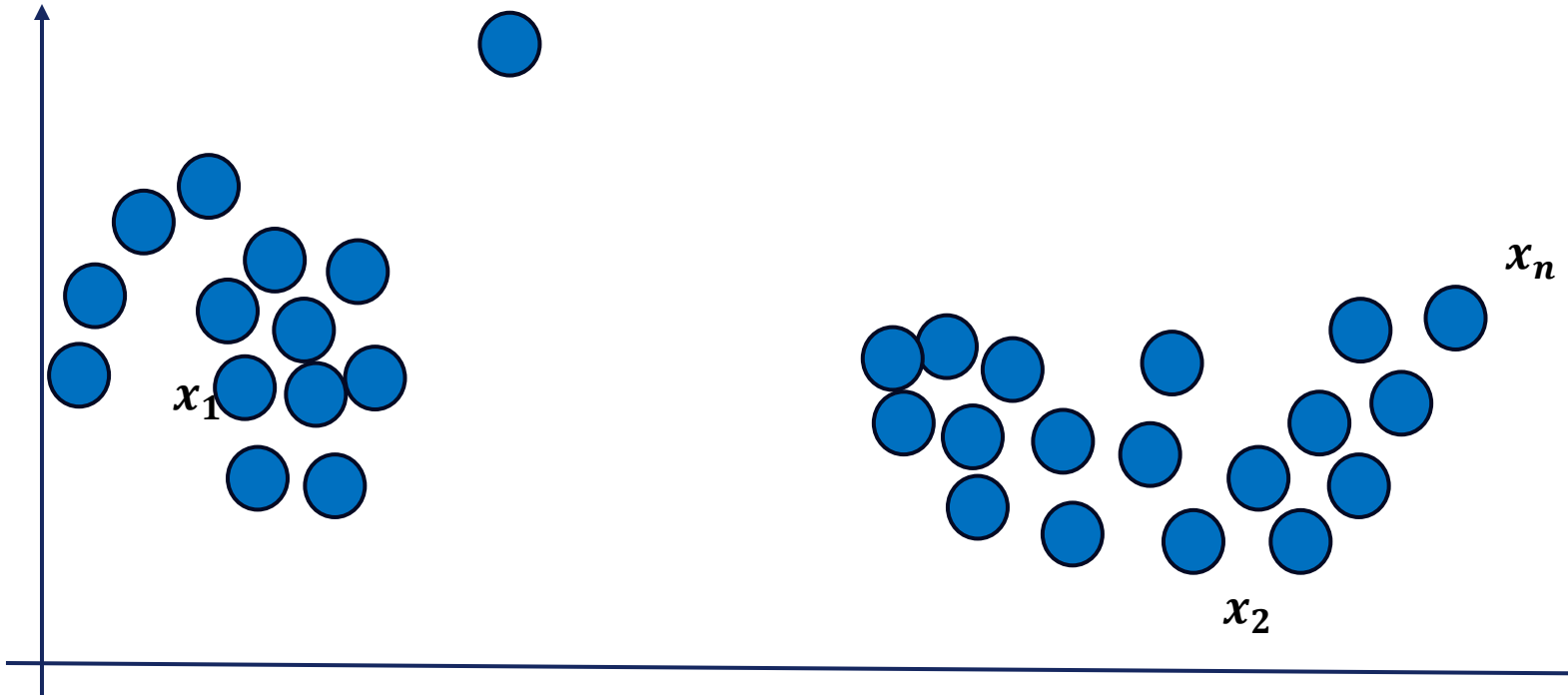
Spectral or subspace clustering methods:

Find a lower dimensional space that better represents the clusters



Clustering techniques

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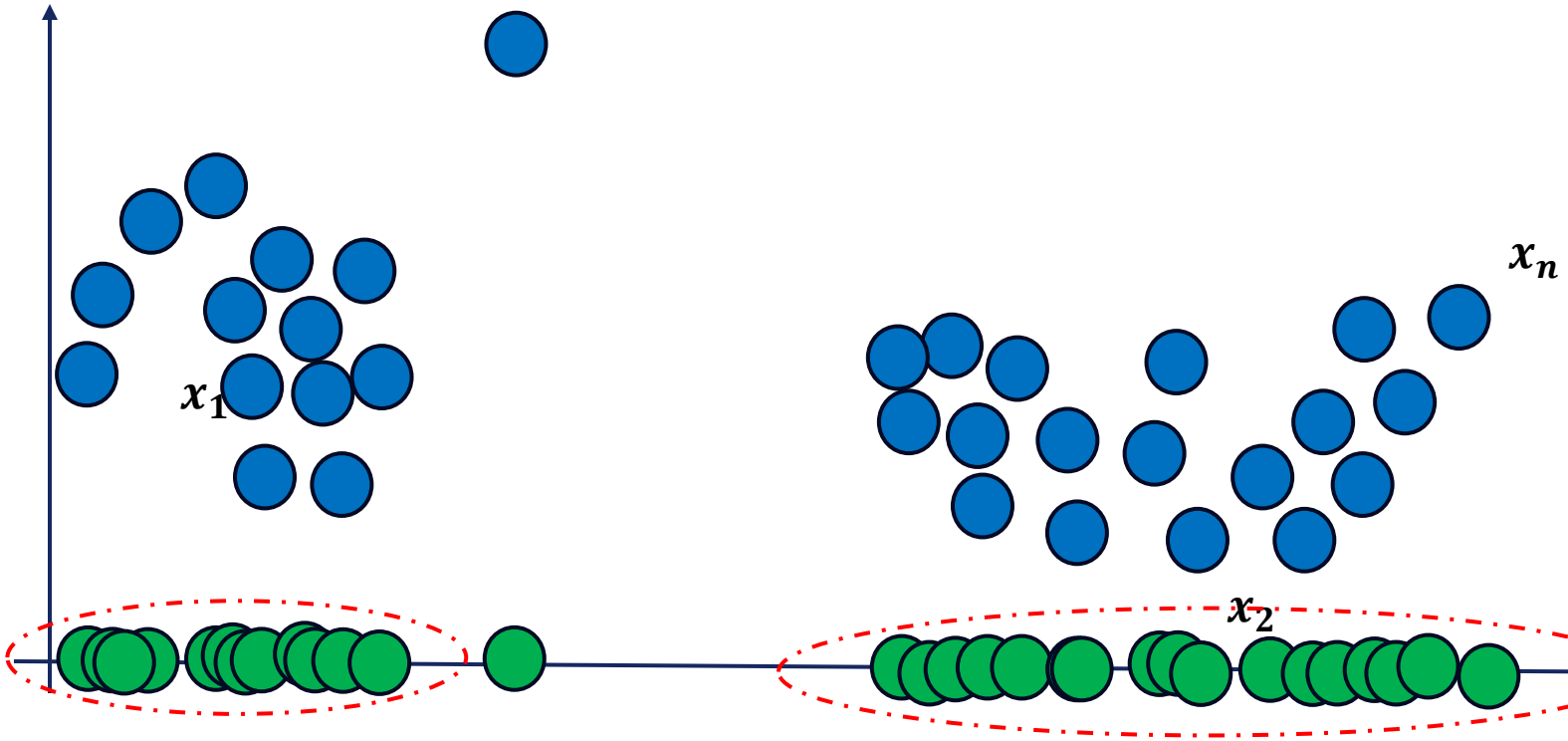

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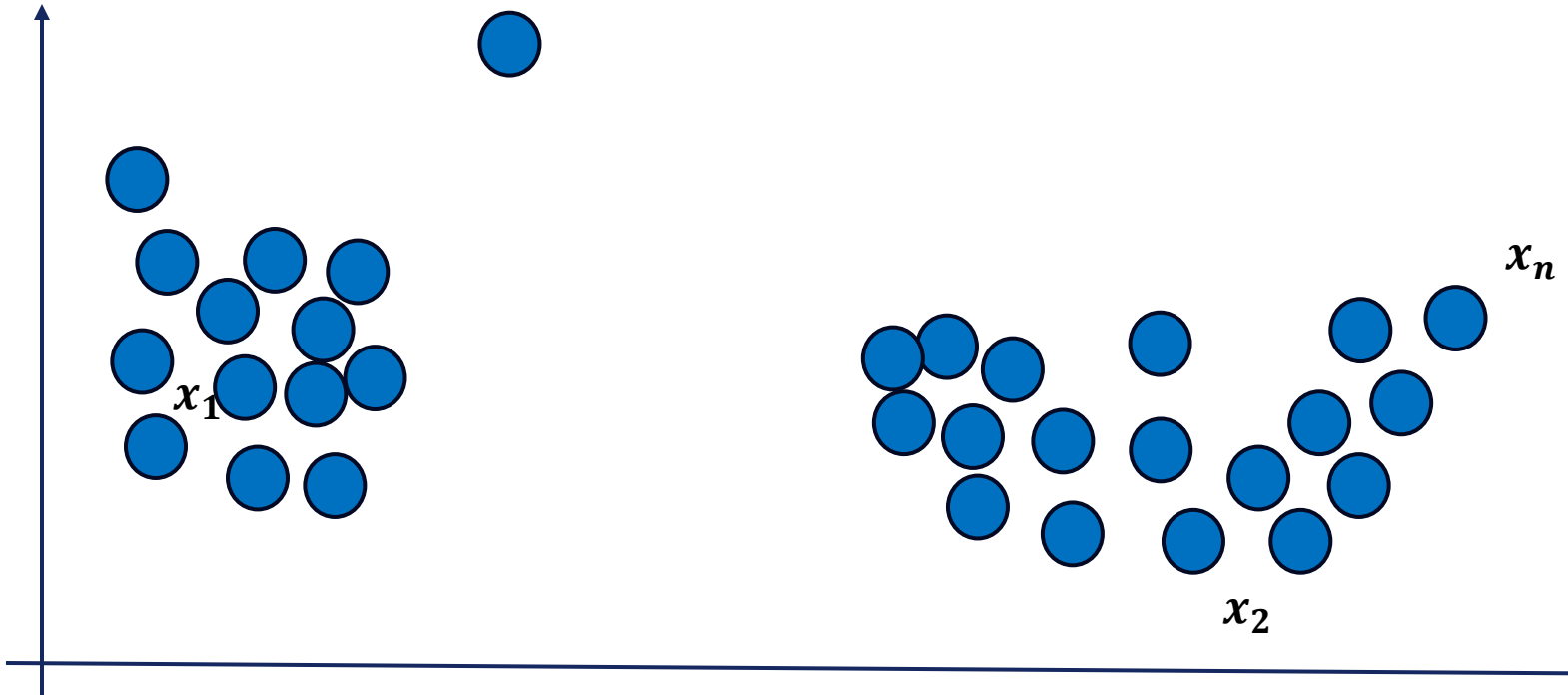
Clustering technique

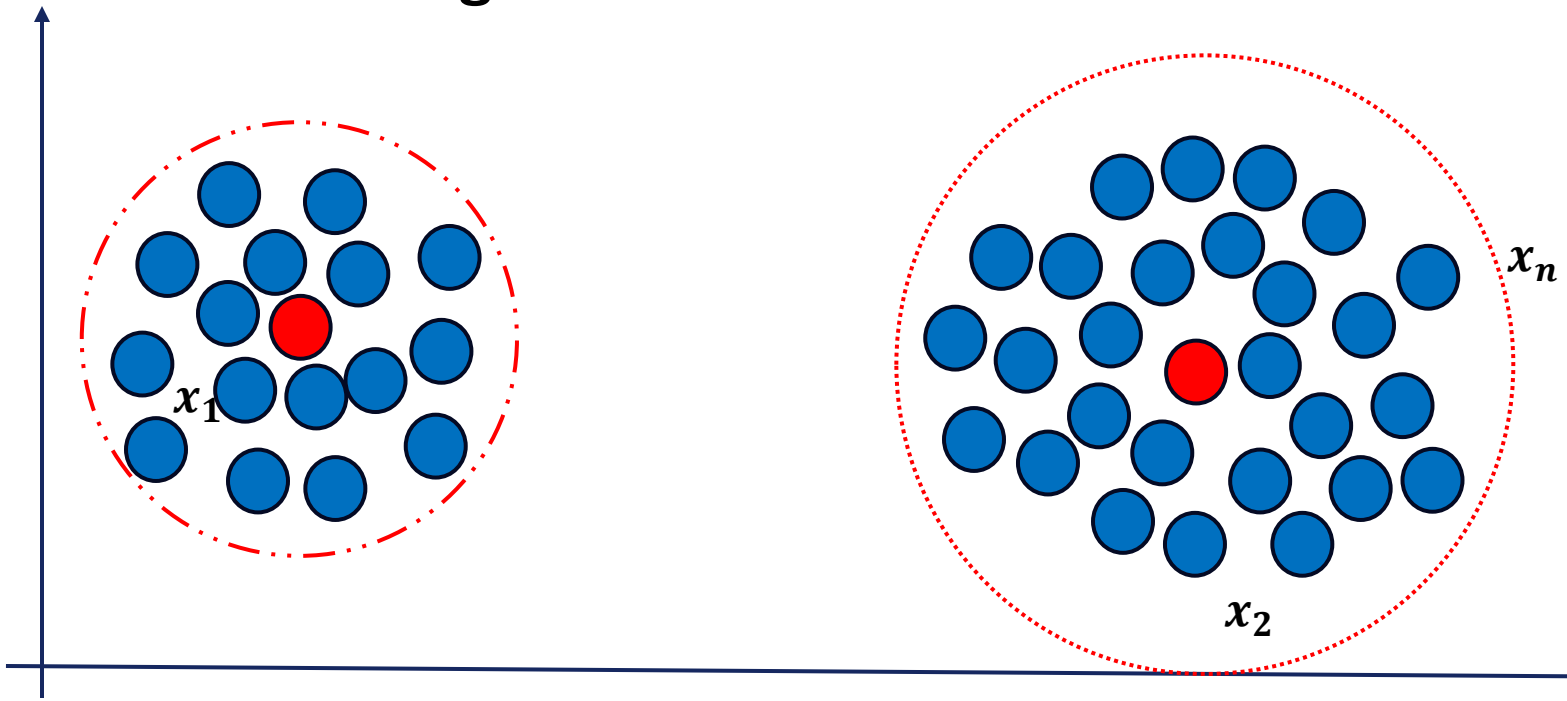
Foundations

- Representative-based method
- Density-based methods
- Hierarchical methods
- Spectral methods
- Graph-based methods

Advanced topics and applications

- Subspace clustering
- Core sets
- Deep learning
- Document clustering
- Clustering for outlier detection



Clustering

- Clustering is broadly and vaguely defined as finding groups of similar entities in a dataset.
- In this class we will learn several clustering techniques and how to validate clustering that we do.
- K-means is a representative-based algorithm that finds a specified number of k of clustering.